# BI2025 Experiment Report – Group 059

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## Abstract

This report documents a machine learning experiment conducted by Group 059 following the CRISP-DM process model. The study focuses on identifying key institutional factors that influence student graduation rates and developing predictive models to support data-driven decision making in higher education.

## CCS Concepts

• **Computing methodologies → Machine learning**.

## Keywords

CRISP-DM, Provenance, Knowledge Graph, Machine Learning, Graduation Rate Prediction, Institutional Performance

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## 1 Business Understanding

### 1.1 Data Source and Scenario

The colleges_usnews dataset used for this project is taken from OpenML. It is a real-world dataset, from the American Statistical Association's 1995 Data Analysis Exposition,based on institutional data published in U.S News and World Report‘s Guide to America‘s Best Colleges (1993-1994). It contains information for 1,302 U.S colleges and universities, including variables related to admission selectivity (SAT/ACT scores, application acceptance), institutional characteristics (public/private, tuition, spending), student composition,faculty qualification and outcomes such as graduation rate. In total the dataset has a total of 35 attributes with most of them being numeric, and some being categorical such as state, public/private. The dataset contains missing values, particularly in the test score variables making it a realistic analysis.

A suitable business scenario for this dataset would be a university administration seeking to understand what factors influence graduation rates. By applying data analytics processes to the dataset, the institution can identify patterns and understand what factors influence a higher graduation rate. This analysis can support strategic planning, resource allocation and performance benchmarking.

### 1.2 Business Objectives

The main objective of this analysis is to define which of the variables have a higher impact on the graduation rate of students. By identifying these factors the institutions have the necessary information to improve students success. Other objectives include providing universities with benchmarks against similar institution, identifying what aspects would have the biggest turn on investments and understanding how admission selectivity or student preparedness relate to graduation outcomes.

All these objectives support organizations in making informed strategic decisions.

### 1.3 Success Criteria

For this project to be called successful, it should produce precise and easily interpretable insights regarding the key factors that influence the graduation rate as well as a model that accurately predicts the graduation rate of an institution based on the provided features. Another measure of success for this analysis would be that the derived recommendations are actionable, such as identifying which variables universities should influence to improve outcomes.

Additionally, the results should be presented in a way that universities can easily use the results for benchmarking, this way they can easily compare their performance against other educational institutions.

### 1.4 Data Mining Goals

The main data mining goal is to build a predictive model that estimates an institution‘s graduation rate based on the available institutional, financial and student-related variables in the dataset. The model should provide accurate predictions, as well as highlight which features contribute most strongly to higher or lower graduation rates. Another goal is to perform exploratory analysis to better understand the relationships between variables such as tuition, selectivity, spending per student, or faculty resources and the graduation rate.This supports the business objective of identifying concrete levers that universities can influence to improve student success.

### 1.5 Data Mining Success Criteria

From a data mining perspective, the project is considered successful if we can train a regression model that generalizes well and explains an important part of the variance in graduation rates. We are trying to have a model that achieves a high coefficient of determination and a low prediction error when testing via cross-validation.In addition, the learned model and derived feature importance measures should be interpretable for non-technical stakeholders. Key factors

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influencing graduation rate should be distinctly identifiable (similar across different validation splits), so that they can be translated into recommendations and benchmarking indicators for universities.

### 1.6 AI Risk Aspects

This project uses institutional-level, historical data about colleges and universities. While the dataset does not contain personal student identifiers, there are still some AI-related risks to consider. First, the data may reflect existing structural inequalities (between public and private institutions or between different regions). A model trained on such data could unintentionally reinforce these patterns if its predictions are used for funding decisions, rankings or policy making without careful interpretation. Second, there is a risk of over-interpreting correlations as causal effects and using the model as a black-box decision tool rather than as analytic support. To handle these risks, the focus of the project is on transparency and being able to interpret the data (using explainable models and feature importance analysis), precisely communicating the limitations of the data and the model, and treating the results as decision-support for experts rather than automatic decision-making.

## 2 Data Understanding

### 2.1 Dataset Description

The dataset contains U.S. colleges and universities derived from US News and World Report. Each row represents one institution and the columns describe selectivity, resources, financial indicators, and student outcomes such as graduation rate.

### 2.2 Raw Data Features

The data set contained the attributes mentioned in the table below:

**Table 1: Raw Data Features**

| Feature | Type | Description |
|---|---|---|
| Additional_fees | numeric | Mandatory non-tuition fees. |
| Average_ACT_score | numeric | Mean ACT score of admitted students. |
| Graduation_rate | numeric | Percentage of students graduating. |
| Public/private_indicator | categorical | Institution control type. |
| State | categorical | Institution location (U.S. state). |
| Student/faculty_ratio | numeric | Students per faculty member. |
| Pct.alumni_who_donate | numeric | Alumni donation rate. |
| Instructional_expenditure_per_student | numeric | Instructional spending per student. |
| Third_quartile-Math_SAT | numeric | 75th percentile Math SAT score. |
| Average_Math_SAT_score | numeric | Mean Math SAT score. |
| Average_Verbal_SAT_score | numeric | Mean Verbal SAT score. |
| Average_Combined_SAT_score | numeric | Mean combined SAT score. |
| First_quartile-Math_SAT | numeric | 25th percentile Math SAT score. |
| First_quartile-Verbal_SAT | numeric | 25th percentile Verbal SAT score. |
| Third_quartile-Verbal_SAT | numeric | 75th percentile Verbal SAT score. |
| First_quartile-ACT | numeric | 25th percentile ACT score. |
| Third_quartile-ACT | numeric | 75th percentile ACT score. |
| Number_of_applications_received | numeric | Total applications received. |
| Number_of_applicants_accepted | numeric | Total applicants accepted. |
| Number_of_new_students_enrolled | numeric | Newly enrolled students. |

### 2.3 Descriptive Statistics

The descriptive statistics for all attributes were computed during the Data Understanding phase. The Table 2 shows a summary of all the data characteristics for the key variables.

**Table 2: Descriptive Statistics of Key Variables**

| Variable | Count | Mean | Std. | Min–Max |
|---|---|---|---|---|
| Public/private indicator | 1302 | 1.64 | 0.48 | 1–2 |
| Average Math SAT score | 777 | 506.8 | 67.8 | 320–750 |
| Average Verbal SAT score | 777 | 461.2 | 58.3 | 280–665 |
| Average Combined SAT score | 779 | 968.0 | 123.6 | 600–1410 |
| Average ACT score | 714 | 22.1 | 2.6 | 11–31 |
| Applications received | 1292 | 2752 | 3542 | 35–48094 |
| Applicants accepted | 1291 | 1871 | 2251 | 35–26330 |
| New students enrolled | 1297 | 779 | 885 | 18–7425 |
| Pct. new students top 10% HS | 1067 | 25.7 | 18.3 | 1–98 |
| In-state tuition | 1272 | 7897 | 5348 | 480–25750 |

### 2.4 Data Quality Assessment

A systematic evaluation of data quality was conducted. Missing values were assessed across all attributes, revealing that 31 variables contain at least one missing observation (see Table 3). In addition, potential outliers were investigated using the interquartile range (IQR) rule, identifying values lying outside the interval $[Q_1 - 1.5 \cdot IQR,\ Q_3 + 1.5 \cdot IQR]$, based on the descriptive statistics reported in Table 2 and in Table 4 we can see the list of outliers.

**Table 3: Number of Missing Values per Attribute**

| Attribute | Missing Values |
|---|---|
| Third quartile ACT | 639 |
| First quartile ACT | 639 |
| Average ACT score | 588 |
| First quartile Math SAT | 530 |
| First quartile Verbal SAT | 530 |
| Third quartile Math SAT | 530 |
| Third quartile Verbal SAT | 530 |
| Average Math SAT score | 525 |
| Average Verbal SAT score | 525 |
| Average Combined SAT score | 523 |
| Board costs | 498 |
| Room costs | 321 |
| Additional fees | 274 |
| Pct. new students top 10% HS | 235 |
| Pct. alumni who donate | 222 |
| Pct. new students top 25% HS | 202 |
| Estimated personal spending | 181 |
| Graduation rate | 98 |
| Room and board costs | 76 |
| Estimated book costs | 48 |
| Instructional expenditure per student | 39 |
| Part-time undergraduates | 32 |
| Pct. of faculty with Ph.D.s | 32 |
| Pct. of faculty with terminal degree | 30 |
| In-state tuition | 30 |
| Out-of-state tuition | 20 |
| Applicants accepted | 11 |
| Applications received | 10 |
| New students enrolled | 5 |
| Full-time undergraduates | 3 |

**Table 4: Outliers for Selected Attributes**

| Attribute | Missing Values |
|---|---|
| Full-time undergraduates | 124 |
| Applications received | 121 |
| Part-time undergraduates | 119 |
| Applicants accepted | 105 |
| New students enrolled | 103 |
| Estimated book costs | 95 |
| Instructional expenditure per student | 84 |
| Pct. new students top 10% HS | 60 |
| Additional fees | 55 |
| Room costs | 47 |
| First quartile Verbal SAT | 35 |
| Average Verbal SAT score | 28 |
| Estimated personal spending | 24 |
| Student/faculty ratio | 22 |
| Average Combined SAT score | 20 |

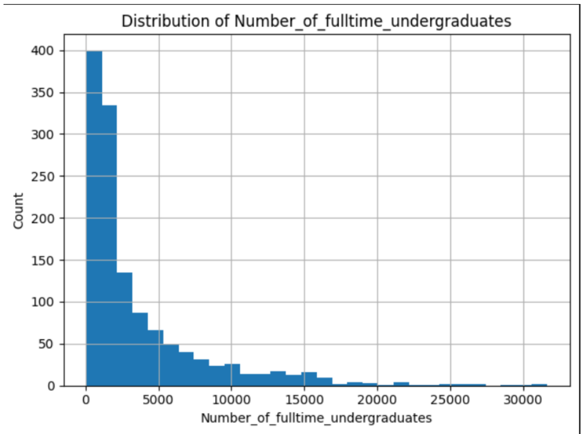


**Figure 1: Distribution of full-time undergraduates**

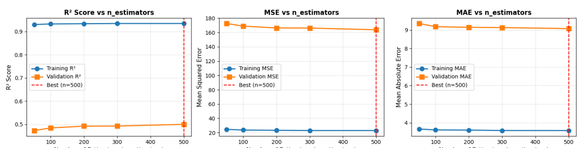


**Figure 2: Full-time undergraduates VS public/private indicator**

## 2.5 Visual Exploration

Simple visual analytics were generated to understand the distribution of the target variable and its relationships with selected numeric features. Summary from the visual exploration activity: The distribution in Fig 1 is highly right-skewed; this is because, Fig 1 shows that, most institutions have a small number of students, with most of them having fewer than 5,000 students. Other institutions exceed this number, with some of them having around 20,000-30,000 students enrolled. From Fig 2 we see that public institutions enroll more students, while private institutions tend to be much smaller. These public institutions appear as outliers but removing them would discard valuable information. We take into account that this may introduce heteroscedasticity and affect linear assumptions, but the model remains suitable for exploratory analysis.

## 2.6 Ethical and Sensitive Aspects (2e)

Potential sensitive attributes include variables related to location, selectivity, or resource levels that might indirectly give us socio-economic status. Analyses and models based on this data risk reproducing existing inequalities between different types of colleges if these attributes are interpreted uncritically as indicators of quality.

Risks include: (1) reinforcing biases embedded in the original ranking methodology (favoring high-resource institutions); (2) misleading stakeholders if predicted graduation rates are treated as objective truth without considering context and (3) drawing causal conclusions from observing aggregate data. Questions for domain experts would be: Which variables are acceptable to use for decision-making? Are there known reporting biases or problematic indicators in the underlying US News data?

Likely required data preparation steps: handling missing values (distinguishing between truly missing vs. 'not applicable'), implementing categorical variables appropriately, considering transformations for skewed financial variables (log-transforming expenditures or tuition) and analysing the impact of extreme outliers (very high-cost or very low-graduation-rate institutions) on model stability.

**Table 5: Dataset Preview (Academic Profile and Selectivity — First 5 Institutions)**

| St. | Type | M-SAT | V-SAT | C-SAT | ACT | Q1-M | Q3-M | Q1-V | Grad. |
|---|---|---|---|---|---|---|---|---|---|
| AK | Pub. | 490 | 482 | 972 | 20 | 440 | 530 | 430 | 15 |
| AK | Priv. | 505 | 460 | 964 | 22 | 460 | 580 | 410 | 39 |
| AL | Priv. | 505 | 460 | 964 | 17 | 460 | 580 | 410 | 40 |
| AL | Pub. | 505 | 460 | 964 | 20 | 460 | 580 | 410 | 55 |
| AL | Priv. | 505 | 460 | 964 | 21 | 460 | 580 | 410 | 51 |

**Table 6: Dataset Preview (Financial, Faculty, and Enrollment Attributes — First 5 Institutions)**

| Board | Add. | Books | Pers. | %PhD | %Term | Stud./Fac. | Alum. | Instr. |
|---|---|---|---|---|---|---|---|---|
| 2500 | 130 | 800 | 1500 | 76 | 72 | 119 | 2 | 10922 |
| 2250 | 34 | 500 | 1162 | 39 | 51 | 95 | 19 | 9584 |
| 1442 | 155 | 500 | 850 | 53 | 53 | 143 | 19 | 7043 |
| 1700 | 300 | 350 | 1248 | 52 | 56 | 328 | 19 | 3971 |
| 2000 | 124 | 300 | 600 | 72 | 72 | 189 | 8 | 5883 |

## 3 Data Preparation

In the data preparation phase we applied several preprocessing steps to ensure that the data is suitable for modeling. First, all the missing values were removed from the graduate_rate variable, as these records can not be used in supervised learning. In the other variables, the missing attributes were replaced using the median of each respective variable, a process that limits the skewed distributions and extreme values. Even though media imputation was applied to obtain an initial clean dataset for exploration this step will be repeated inside the modeling pipeline only for the training data to avoid data leakage. After median imputation, public/private institutional indicator was converted from numeric codes to meaningful category labels. Lastly, categorical attributes were converted to a category data type so that they can be correctly handled in the future modeling steps. These preprocessing actions resulted in a cleaned dataset with complete numeric information and clearly defined categorical variables, while preserving the original semantic structure of the data. Table 5 and Table 6 show an overview of the cleaned data.

*3.0.1 Other preprocessing steps considered.* During the data preparation phase, several additional preprocessing actions were evaluated but ultimately not applied.

- **Outlier removal.** An outlier analysis based on the IQR rule revealed a substantial number of extreme values in several numeric attributes, particularly `Number_of_fulltime_undergraduates` (124 outliers), `Number_of_applications_received` (121 outliers), and `Number_of_applicants_accepted` (105 outliers), as well as

in cost-related variables. Visual inspection using scatter plots by institution type indicated that these values correspond to large public universities or highly selective private institutions. Removing these observations would therefore eliminate meaningful structural variation, and outliers were retained.
- **Feature scaling.** Feature scaling was considered but postponed to the modeling phase. Only certain regression algorithms (e.g., linear regression, support vector regression, and neural networks) require standardized inputs, while tree-based models are scale invariant. Scaling of SAT scores and cost-related variables will therefore be applied conditionally, depending on the selected algorithm.
- **Encoding of high-cardinality categorical variables.** Encoding attributes with many distinct values, such as `State`, would significantly increase the dimensionality of the dataset. This step was deferred to the modeling phase and will be applied only if the chosen algorithm requires numeric input.
- **Attribute removal.** The removal of attributes with many missing values or limited variance such as the SAT and ACT scores was considered but rejected, as all variables carry relevant semantic information about institutional characteristics.

*3.0.2 Possible derived attributes.* Based on the exploratory analysis, several derived attributes were identified as potentially useful for modeling.

- A *selectivity ratio*, defined as `Number_of_applications_received` divided by `Number_of_applicants_accepted`, to capture institutional selectivity.
- A *total cost* variable, defined as the sum of `In_state_tuition`, `Room_and_board_costs`, `Additional_fees`, `Estimated_book_costs`, and `Estimated_personal_spending`, representing the overall annual cost of attendance.

These derived features may be incorporated during the modeling phase if they prove useful for predicting graduation rates.

*3.0.3 External data sources.* Graduation outcomes may also be influenced by factors beyond an institution's internal characteristics. Potential external data sources include state-level socio-economic indicators such as median household income, unemployment rates, and cost-of-living indices. Incorporating such data would allow tuition and expense variables to be normalized across regions. Additionally, high-school graduation and college enrollment rates at the state level could provide context on student preparedness, which may further affect graduation outcomes.

# 4 Data Modeling

## 4.1 Identifying suitable data mining algorithms

To predict the graduation rates for this dataset, we considered several algorithms:

1. Linear Regression: Simple, interpretable, but may not capture non-linear relationships
2. Random Forest Regression: Handles non-linear relationships, provides feature importance, robust to outliers
3. Gradient Boosting (XGBoost/GBM): Often achieves high performance but more complex and computationally expensive
4. Support Vector Regression: Can handle non-linear relationships but less interpretable

After carefull consideration we chose Random Forest Regression as the primary algorithm because first, it handles mixed data types well without extensive preprocessing. Second it provides feature importance for interpretability. Third is robust to outliers and missing values. Fourth it provides a good balance between performance and interpretability. Fifth it can capture non-linear relationships between features and graduation rate. Finally is less prone to overfitting than single decision trees.

## 4.2 Identifying hyperparameters for tuning

The Random Forest algorithm provides several hyperparameters that can be tuned to provide the desired results. One of the key parameters is n_estimators which shows the number of trees in the forest; the more trees, the better the results but the slower the processing. Another important parameter is max_depth which shows the maximum depth of trees. Additionally, min_sample_split shows the minimum samples required to split a node, while min_sample_leaf shows the minimum samples required at a leaf node. Finally, max_features parameter shows the number of features to consider in order to get the best split.

The parameter that we decided to choose is n_estimator because it has a significant impact on model performance, it's computationally efficient to tune because it can test multiple values, and it's less likely to cause overfitting especially compared to max_depth. Moreover it provides a good balance between model complexity and training time

To decide the optimal number of trees, the following values were tested: [50, 100, 200, 300, 500].

## 4.3 Defining Train/Validation/Test splits

To split the data into train/validation/test sets, we first defined a random number split with a random_state of 42 for reproducibility. Then we defined that the test set will be 15%, and it was held out for final evaluation only. Additionally, the validation set was also defined to be 15% to be used for hyperparameter tuning. Leaving the training set with 70%.

This 70/15/15 split was chosen to: 1. Provide more training data (70%) to help reduce overfitting 2. Ensure adequate validation data (15%) for reliable hyperparameter selection 3. Maintain an independent test set (15%) for unbiased final performance assessment 4. Ensure reproducibility through fixed randomy_state

There was no need for stratification, as this regression problem has a continues target; also, because there where no time dependencies identified in the data, the random split is appropriate.

## 4.4 Model Training

The Random Forest models were trained with n_estimators values: [50, 100, 200, 300, 500]. Each model was trained on the training set and evaluated on both training and validation sets. Performance metrics ($R^2$, MSE, MAE) were computed for each configuration the results are documented at Table 7. The tuning process shows that performance generally improves with more trees, but with diminishing returns.

**Table 7: Random Forest Hyperparameter Tuning Results**

| n_estimators | Train $R^2$ | Val. $R^2$ | Val. MSE | Val. MAE |
|---|---|---|---|---|
| 50 | 0.9303 | 0.4729 | 172.54 | 9.35 |
| 100 | 0.9331 | 0.4843 | 168.82 | 9.17 |
| 200 | 0.9341 | 0.4920 | 166.30 | 9.14 |
| 300 | 0.9351 | 0.4924 | 166.15 | 9.12 |
| 500 | 0.9351 | **0.4998** | **163.73** | **9.07** |

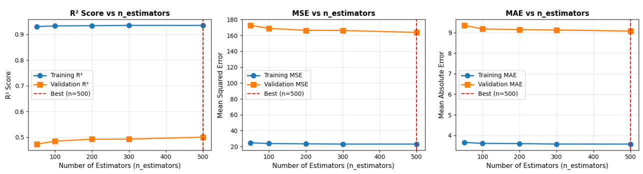


**Figure 3: Random Forest Hyperparameter Tuning Results**

## 4.5 Performance Metrics

The most suitable model according to our results is the one with n_estimators configuration of 500, as we can see from Fig 3 This model explains approximately 50% of the variance in the validation dataset. In addition it achieves a validation MSE of 163.73 and a validation MAE of 9.07, indicating improved performance compared to other models.

## 4.6 Retraining the model

During the validation phase, the ideal hyperparameter configuration was identified, and the final Random Forest model was retrained with the combined training and validation datasets. This step conforms to standard machine learning practice and seeks to maximise the quantity of data available for learning before assessing the model on the independent test set.

Specifically, the training and validation sets were combined, yielding a total of 1,023 samples for final model training. To guarantee methodological consistency, the model was re-trained with the previously selected optimal hyperparameter value of `n_estimators = 500`, while all other parameters remained constant from the tuning phase.

To ensure successful learning, the re-trained model's performance was tested on the entire training data set. The final model had a $R^2$ of 0.9356, an MSE of 22.56, and an MAE of 3.49. As expected, these metrics show a strong fit to the combined training data, reflecting the increased quantity of information available to the model.

This re-trained model is used as the final predictive model, and it is then tested on the remaining test set to get a fair estimate of its generalisation performance.

# 5 Evaluation

During evaluation, the final Random Forest model (estimators = 500) was first retrained on the combined training and validation sets, and then applied exactly once to the test set. This follows standard practice to maximize the data available for learning, but also getting an unbiased estimate of generalization performance. The retrained model achieved very strong performance on the combined training data ($R^2$ = 0.936, MSE = 22.56, MAE = 3.49), which is expected, because the model has already seen these observations during training. The more informative picture comes from the independent test evaluation.

On the held-out test set, the final model obtained a mean absolute error (MAE) of 10.78, a root mean squared error (RMSE) of 14.60, a mean squared error (MSE) of 213.29 and a coefficient of determination $R^2$ of 0.456. Since graduation rate is measured in percentage points, an MAE of around 10.8 means that, on average, the model's prediction deviates by roughly eleven percentage points from the true graduation rate. An $R^2$ value of approximately 0.46 indicates that the model explains about 46% of the variance in graduation rates across institutions. This represents a moderate but meaningful level of predictive power in a noisy, real-world dataset with substantial missing and diverse value among institutions.

To provide a compact overview and to satisfy the requirement of comparing against a trivial baseline, Table 8 summarizes the test performance of the final Random Forest model and of a naive predictor that always outputs the mean graduation rate observed in the training data.

**Table 8: Test-Set Performance: Final Model vs. Trivial Baseline**

| Model | MAE | RMSE | MSE | $R^2$ |
|---|---|---|---|---|
| Random Forest (final) | 10.78 | 14.60 | 213.29 | 0.456 |
| Mean baseline | 16.25 | 19.80 | 391.96 | $\approx$ 0.00 |

The table shows that if compared to the trivial mean predictor, the Random Forest regressor clearly improves performance: the MAE is reduced by about 5.47 percentage points (from 16.25 to 10.78), and the MSE is almost halved (from 391.96 to 213.29). At the level of explained variance, the baseline accounts for essentially none of the variation in graduation rates, whereas the final model achieves $R^2 = 0.456$. This shows that the model captures substantial structure in the data beyond a constant prediction and achieves the requirement to compare against a trivial baseline.

To address the requirement of identifying state-of-the-art performance, we searched for published benchmark studies using the chosen dataset in particular for regression tasks predicting graduation rate. We did not find peer-reviewed work reporting comparable numeric results for this specific prediction task. The dataset appears to be used mainly for teaching, exploratory data analysis and demonstration purposes rather than as a standard benchmark for graduation-rate regression. In the absence of external numeric baselines, we treat our tuned Random Forest model as an internal benchmark and focus on the comparison to the trivial mean predictor and to the success criteria defined in the Business Understanding phase.

From a business and data-mining perspective, the success criteria required that the model should explain a substantial fraction of the variance in graduation rates and achieve a reasonably low prediction error, while remaining interpretable enough to support decision-making. With an $R^2$ of 0.456 on the test set and an MAE of 10.78 percentage points, the model partially fills these criteria. It does significantly better than a naive baseline and highlights relevant predictors of graduation rate, but it does not achieve perfect accuracy, which is realistic given the complexity of the underlying process and the limitations of the available data. Overall, the results are consistent with the expectations for this type of institutional-level dataset and are sufficient to support exploratory, decision-support use rather than high-stakes automated decision-making.

To evaluate potential bias, the `Public/private_indicator` was selected as a "protected" attribute that distinguishes between public and private institutions. Using the original cleaned dataset, we computed error metrics separately for both groups on the test set. For private institutions ($n = 59$), the model achieved MAE = 9.29, RMSE = 11.48 and $R^2 = 0.511$. For public institutions ($n = 122$), the corresponding metrics were MAE = 11.51, RMSE = 15.90 and $R^2 = 0.341$. Thus, the model is more accurate and explains more variance for private colleges than for public ones. The difference in MAE of about 2.2 percentage points and the gap in $R^2$ suggests a moderate performance difference between these groups.

This performance difference can have several explanations. On the one hand, private institutions may have more homogeneous characteristics (e.g., tuition level, selectivity, spending per student), which makes their graduation rates easier to model. Public institutions, by contrast, cover a much wider range of sizes, missions and funding structures, which may lead to greater variability that the current model cannot fully capture. On the other hand, the observed difference also indicates a potential bias in the model's error distribution: predictions for public institutions tend to be less accurate than for private ones. A more detailed fairness analysis, including explicit fairness metrics and consultation with domain experts, would be necessary before using the model in any context where such differences could lead to disadvantageous treatment of a particular institutional subgroup.

In summary, the evaluation phase shows that the final Random Forest model achieves a clear and substantial improvement over a trivial baseline and provides a reasonable predictive approximation of graduation rates, while still exhibiting moderate generalization errors and unequal performance across institutional types. These findings underline that the model is suitable as an analytical and decision-support tool, but not as an automated decision-making system. They also motivate potential future work on model refinement, such as incorporating additional external data sources, experimenting with alternative algorithms, or addressing fairness constraints to reduce performance differences between public and private institutions.

# 6 Deployment

## 6.1 Comparison and Recommendations

The evaluation results show that the final Random Forest model partially meets the business success criteria defined in the business understanding phase. On the one hand, the model achieves a clear improvement over a trivial mean baseline on the test set (MAE reduced from 16.25 to 10.78 percentage points, MSE almost halved, and $R^2$ increasing from approximately 0 to 0.456). This indicates that the model captures substantial structure in the data and explains a meaningful part of the variance in graduation rates.

On the other hand, the remaining prediction error of around eleven percentage points MAE, and an $R^2$ clearly below 1.0, show that the model is far from perfectly accurate. Given the complexity and noisiness of institutional data, this is not unexpected, but it implies that the model should not be used as the sole basis for high-stakes decisions (such as funding allocation or detailed performance evaluation of individual institutions). Instead, it is better suited as an analytical support tool to identify patterns, compare institutions at an aggregated level, and explore potential scenarios, consistent with the business objectives of benchmarking, understanding factors of graduation rate and informing strategy rather than fully automating decisions.

We recommend a hybrid decision-support setup. The model could be integrated into a dashboard where institutional analysts can inspect predicted graduation rates, feature importance, and group performance. If a more detailed deployment is desired, it would be reasonable to restrict automated usage to parts of the data space where model performance is known to be stable (for example, institution types and size ranges with sufficient training data), while flagging predictions in more uncertain regions for manual review. Further analyses that would strengthen deployment include: (i) experimenting with alternative algorithms and feature engineering (e.g. derived selectivity and total cost measures), (ii) integrating external socio-economic data to improve explanatory power, and (iii) performing a more systematic fairness and robustness analysis across additional subgroups.

## 6.2 Ethical Aspects

The deployment of a predictive model for graduation rates involves several ethical aspects that build directly on the AI risk considerations formulated in the business understanding phase and the sensitivity analysis in the data understanding phase. First, many variables in the data,set (such as public vs. private status, spending per student, or selectivity measures) reflect structural inequalities between institutions. If model outputs were used without a prior critical analysis for rankings, funding decisions, or public communication, there is a risk of reinforcing existing advantages of institutions that are well rescoured and penalizing those with less preferable conditions. The evaluation phase also showed moderate performance differences between public and private institutions, with better accuracy for private colleges, which may further indicate unequal treatment if not explicitly addressed.

Second, there is a danger of treating predicted graduation rates as objective quality indicators. The model is trained on historical data and correlations. It does not capture all relevant factors (such as regional labour markets, social and economic background of students, or institutional missions) and does not provide causal explanations. Using such predictions to make significant decisions (closing programmes, changing admissions policy, or reallocating funding) without careful expert review would be ethically problematic. The earlier analysis in 1.f) and 2.e) already highlighted the risk of over-interpreting correlations as causal and of embedding the original US News ranking biases into downstream decision-making.

In light of these risks, deployment should explicitly frame the model as a decision-support tool rather than an automatic decision system. Predicted values and feature importance measures should be communicated together with uncertainty and limitations, and stakeholders should be cautioned against interpreting them as definitive judgements of institutional quality. Any use of the model in policy or funding contexts should be accompanied by transparency about the data sources, modeling assumptions, and known fairness limitations, and should include mechanisms for contesting or overriding model-based suggestions, especially in cases where disadvantaged institution types might be negatively affected.

### 6.3 Monitoring Plan

To ensure the continued reliability and ethical use of the deployed graduation rate prediction model we need to ensure monitoring of the predictive performance, data drift and fairness across key subgroups. To do this, first the model performance on newly arriving data should be tracked using the same metrics as in the evaluation phase (MAE,RME and $R^2$). We can evaluate how the model is performing by comparing the future metrics with the baseline established during the current experiment. If we notice a significant degradation on these metrics, that would indicate that the relationship between predictors and graduation rates has changed which means that there is a concept drift or that quality issues have emerged in the data.

Second, we need to monitor the input data distribution needs. Which means tracking basic, descriptive statistics and missing-value patterns for key features such as SAT/ACT scores, cost variables, and enrollment numbers. Furthermore, we need to monitor changes in the percentage of public vs. private institutions or different size classes. By investigating these aspects for large distributions shifts or sudden changes in the frequency of missing values, we make sure that the model remains applicable or whether retraining and possible re-design of preprocessing steps are required.

Third, fairness and bias aspects should be monitored over time. Error metrics consisting of MAE, RMSE, $R^2$ should be calculated separately for each new evaluation period. If over time from investigating these values we exam that the difference in errors between these groups becomes too large, such as MAE differing by more than a few percentage points, this means that the features used for this model and how the predictions are applied in decision making need a closer review.

Additionally, it is important to record cases where model's predictions are overridden by human experts and whether these overrides follow clear patterns. These pattern can help indicate weaknesses in the model or ethical concerns.

To conclude these monitoring activities provide clear triggers for intervention, such as retraining the model, adjusting the feature set, limiting the model's use to certain cases, or, in extreme situations, temporarily stopping the use of the model until the issues are better understood.

### 6.4 Reproducibility

This experiment was developed with reproducibility in mind, and this can be seen in many factors, such as the use of a publicly available dataset (colleges_usnew), a clearly defined conda environment with an requirements file and fixed number of seeds which was used for data splitting and model training. The whole workflow is implemented in a notebook with clearly separated steps for data loading, data preprocessing, modeling and evaluation. Furthermore, the provenance knowledge graph records which activities were carried out, when they happened, and which data, models and results were involved.

With all that being said this experiment has also some limitations regarding reproducibility. Reproducibility depends on the continued availability and stable behaviour of external services such as OpenML and the provenance database. Another factor that can impact reproducibility is changes is software versions, for example changes in Python or scikit-learn may lead to slightly different numerical results. Furthermore some parts of the Data Understanding and Data Preparation phases required manual inspection, such as identifying outliers and deciding not to remove them and even though these decisions are documented they introduce a degree of subjectivity that cannot be fully automated.

All in all, we can confidently say that the experiment is reproducible in terms of its main steps and conclusions. If another analyst follows the documented process and uses the same dataset with similar software versions should obtain comparable results and insights. With that being said, exact reproduction of every numeric result cannot be fully guaranteed based on the report and provenance information alone, because details such as exact library versions, hardware, and the state of external services are not fully controlled.

## 7 Conclusion

Working through the full CRISP-DM cycle on a real-world dataset highlighted how strongly the different phases of a data analytics project depend on each other . Early choices in the Business Understanding and Data Understanding phases, such as defining graduation rate as the primary target and success criteria, directly impacted preprocessing, modeling, and evaluation. Having clear criteria early helped us interpret the final results, where the model achieved moderate predictive performance with an $^2$ 0.46 and an MAE 10.8. These results show that the model is appropriate for exploratory decision support rather than automated decision-making.

From a modeling perspective, the experiment demonstrated both the strengths and limits of Random Forests on noisy, heterogeneous institutional data. Although the final model clearly outperformed a trivial baseline and captured meaningful structure in the data, a substantial portion of the variance in graduation rates remained unexplained, underlining the complexity of the underlying process.

From a technical and organizational perspective, the exercise highlighted the importance of reproducibility and structured documentation. Using fixed data splits, a controlled environment, and consistently logging activities and entities into the provenance knowledge graph all added overhead but also increased transparency and made the workflow easier to understand and reproduce. Finally, evaluating performance across public and private institutions showed that acceptable overall accuracy can still hide subgroup differences, reinforcing the need to consider fairness and ethical aspects alongside predictive performance.

Overall, the exercise reinforced that successful business intelligence projects are not only about achieving good predictive accuracy but also about building transparent, well-documented, and responsible analytical processes that can be understood and critiqued by stakeholders beyond the data science team.